

[illegible]

Figure 1 is a line plot showing the Kinetic Energy Z-score (Envelope) for the 21S rRNA gene. The x-axis represents the nucleotide position (nt) from 18,500 to 22,500. The y-axis represents the Kinetic Z-score from 0 to 15. A dashed red line indicates the Significance Threshold (Z = 3.0). The plot shows several peaks above the threshold, each labeled with its corresponding nucleotide range, weight (w), and $\Delta \ln L$ value.

Peak Label	Nucleotide Range (nt)	Weight (w)	$\Delta \ln L$
[19270, 19290]	19270 - 19290	20	+28.47
nt 19855 [19853, 19857]	19853 - 19857	4	+85.41
nt 20816 [20800, 20831]	20800 - 20831	31	+68.32
nt 21321 [21321, 21321]	21321	0	+22.77
nt 21427 [21420, 21434]	21420 - 21434	14	+17.08

The plot displays genomic alignment coordinates (nt) on the x-axis, ranging from 0 to 35,000 nt. The y-axis lists three methods: RhizAeon (Kinetic Manifold + ML Polishing), RDP4 Bootscanning (Kolb et al. 2024, 1800-bp Window), and RF-Net 2 Phylogenetic Network Reticulations. The plot shows support for various methods across the coordinate range. Key features include:

- RhizAeon (Kinetic Manifold + ML Polishing):** Shows a green box labeled "Hexon (w = 0-20 nt)" and a green arrow pointing to it with the text "RhizAeon Exact Physical Resolution (w = 0-20 nt) Single-base profile likelihood support".
- RDP4 Bootscanning (Kolb et al. 2024, 1800-bp Window):** Shows a red box labeled "Penton window (1,800 nt blurred span)" and a red box labeled "Fiber window (2,400 nt blurred span)". A red arrow points to the red box with the text "RDP4 Window Smearing (1,800 nt width) Precluded single-base breakpoint attribution".
- RF-Net 2 Phylogenetic Network Reticulations:** Shows a purple box labeled "Reticulation 10 (Penton swap)" and a purple box labeled "Reticulation 12/15 (Hexon swap)".

Genomic Alignment Coordinate (nt)

06_adenovirus_bixby2024

Human Mastadenovirus B (HAdV-B)

Bascom Palmer Eye Vitreous Isolates

Kolb et al. 2024 (IOVS)

Modular Mosaicism (Hexon/Fiber)

8 Genomes x 38,452 nt

15,557 Informative Sites

1090353a65f3... (Verified)

95.26 ms (0.095 s)

>1,000× Acceleration

nt 14,916 [14914, 14919] (w=5 nt)

nt 19,280 [19270, 19290] (w=20 nt)

nt 19,855 [19853, 19857] (w=4 nt)

nt 21,321 [21321, 21321] (w=0 nt)

nt 34,767 [34765, 34769] (w=4 nt)

Replaces 1.8 kb RDP4 window
with 0-20 nt physical plateaus